NORMAL INCIDENCE — Γ AND τ

Med. 1 ($z<0$): incident side. Med. 2 ($z \ge 0$): transmitted side. Boundary at $z=0$.

Reflection coeff. *wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Intrinsic impedance *general → nonmag. shortcut*

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r = 1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

Default $\mu_r = 1$ (air, dielectric, "nonmag."). Only use $\mu_r \neq 1$ if problem says so ("ferromagnetic", iron, ferrite, or μ_r given explicitly).

Low-loss η_2 (poor conductor) $\sigma/(\omega\varepsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop tiny } j} \frac{\eta_0}{\sqrt{\varepsilon_{r,2}}}\ (\Omega)$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_{r,2}}$. The j term is for accuracy with $|\eta_c|, \theta_\eta$.

$\tilde{\text{E}}$ PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c} \text{ (rad/m)} \quad \Rightarrow \quad k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table *$u \in \{x, y, z\}$ = axis perp. to boundary*

Inc dir	Inc	Refl	Trans
$+\hat{u}$	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
$-\hat{u}$	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \ge 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$\tilde{\mathbf{E}}^i = a_0 \hat{e}\, e^{\mp jk_1 u}, \quad \tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e}\, e^{\pm jk_1 u}, \quad \tilde{\mathbf{E}}^t = \tau a_0 \hat{e}\, e^{\mp jk}$

(lossy med 2: replace $e^{\mp jk_2 u}$ with $e^{\mp\alpha_2 u} e^{\mp j\beta_2 u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x}+\hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x\hat{x} + a_y e^{j\delta}\hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100\,|\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100\,|\tau|^2\,\frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

LOSSY MEDIA — 5 CASES

Loss tangent $\varepsilon' = \varepsilon_r\varepsilon_0$, $\varepsilon_0 \approx 1/(36\pi \times 10^9)$

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

Type	$\sigma/\omega\varepsilon$	η_c, α, β
Lossless ($\sigma=0$)	0	$\eta = \eta_0/\sqrt{\varepsilon_r}$ exact; $\alpha = 0$, $\beta = \omega\sqrt{\varepsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1+j\frac{\sigma}{2\omega\varepsilon'}\right)$; $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}$, $\beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}$
Quasi-cond.	10^{-2}–10^2	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$, see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f\mu/\sigma}$; $\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\varepsilon_r}\,\eta_0$

For Γ/τ: drop j correction; use $\eta_0/\sqrt{\varepsilon_r}$. Magnetic: keep μ_r.

Exact (quasi-cond) $X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1}, \ \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}, \ \theta_\eta = \frac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}$$

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin\theta_2 = \sin\theta_1\sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin\theta_1\,\frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1 → 2 → 3 → 4*

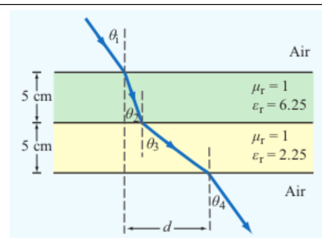
$$\sin\theta_{i+1} = \sin\theta_i\sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs *thickness t_i, refracted angle inside slab i*

Slab-indexed *θ_i = angle inside slab i*

$$d = \sum_{i=1}^{N} t_i\,\tan\theta_i$$

Angle-indexed (HW3 P3 style) *θ_1 = incident, θ_{i+1} = inside slab i*

$$d = \sum_{i=1}^{N} t_i\,\tan\theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes $t_1\tan\theta_2$, slab 2 contributes $t_2\tan\theta_3$. The angle in the formula is always the one the ray makes inside *that slab.*

Air–slab–air: exit angle returns to θ_1 but is laterally shifted by d.

FRESNEL OBLIQUE — Γ, τ, R, T

Perpendicular ($E \perp$ plane of inc.)

$$\Gamma_\perp = \frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t}, \quad \tau_\perp = 1 + \Gamma_\perp$$

Parallel ($E \parallel$ plane of inc.)

$$\Gamma_\parallel = \frac{\eta_2\cos\theta_t - \eta_1\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i}, \quad \tau_\parallel = (1 + \Gamma_\parallel)\frac{\cos\theta_i}{\cos\theta_t}$$

Critical angle (TIR) *only if $n_1 > n_2$*

$$\sin\theta_c = \frac{n_2}{n_1} = \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}} \ \Rightarrow\ \theta_i \ge \theta_c = \text{total internal refl.}$$

Brewster angle ($\parallel$ only) *$\Gamma_\parallel = 0$, no reflection*

$$\tan\theta_{B\parallel} = \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}} \text{ (nonmag.)}$$

Reflectivity / Transmissivity *power form; $R + T = 1$*

$$R = |\Gamma|^2, \quad T = |\tau|^2\,\frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}$$

Normal incidence: $\theta_t = \theta_i = 0$, cos factors drop, $R = |\Gamma|^2$, $T = |\tau|^2(\eta_1/\eta_2)$.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Shorthand: $c_i = \cos\theta_i$, $c_t = \cos\theta_t$.

	Normal ($\theta_i = 0$)	**$\perp$ pol**	**$\parallel$ pol**
Γ	$\frac{\eta_2-\eta_1}{\eta_2+\eta_1}$	$\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$	$\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$
τ	$\frac{2\eta_2}{\eta_2+\eta_1}$	$\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$	$\frac{2\eta_2 c_i}{\eta_2 c_t + \eta_1 c_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel)\frac{c_i}{c_t}$
R	$\lvert\Gamma\rvert^2$	$\lvert\Gamma_\perp\rvert^2$	$\lvert\Gamma_\parallel\rvert^2$
T	$\lvert\tau\rvert^2\frac{\eta_1}{\eta_2}$	$\lvert\tau_\perp\rvert^2\frac{\eta_1 c_t}{\eta_2 c_i}$	$\lvert\tau_\parallel\rvert^2\frac{\eta_1 c_t}{\eta_2 c_i}$
$R+T$	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\sin\theta_i$; $\eta_i = \sqrt{\mu_i/\varepsilon_i}$; nonmag. $\Rightarrow \eta_2/\eta_1 = n_1/n_2$.

FIBER OPTICS

Core index n_f, cladding n_c, surrounded by n_0. Need $n_f > n_c$ for TIR.

Index of refraction *nonmag. $\mu_r = 1$*

$$n = c/u_p = \sqrt{\varepsilon_r}$$

Acceptance angle θ_a *max θ_i for TIR in fiber*

$$\sin\theta_a = \frac{1}{n_0}\sqrt{n_f^2 - n_c^2}$$

Modal dispersion (max time delay) *fiber length ℓ*

$$\tau = \frac{\ell n_f}{c}\cdot\frac{n_f - n_c}{n_c}$$

Max data rate *$T \ge 2\tau$ to avoid pulse overlap*

$$f_p = \frac{1}{2\tau} = \frac{c\,n_c}{2\ell n_f(n_f - n_c)}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0$, $H_z \neq 0$

TM mode $H_z = 0$, $E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE$_{mn}$ vs TM$_{mn}$ *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. "a TE wave" → TE).
2. If $m=0$ or $n=0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).
3. Source field $H_z(x,y)$ given $\Rightarrow$ TE. Source field $E_z(x,y)$ given $\Rightarrow$ TM.

TE allows at least one of $m, n \ge 1$ (other can be 0). TM needs both $m, n \ge 1$. That's why TE$_{10}$ exists but TM$_{10}$ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: $C_X = \cos\frac{m\pi x}{a}$, $S_X = \sin\frac{m\pi x}{a}$, $C_Y = \cos\frac{n\pi y}{b}$, $S_Y = \sin\frac{n\pi y}{b}$. All fields carry an $e^{-j\beta z}$ factor (omitted below). $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$.

TE mode *generator: $\tilde{H}_z$*

$$\tilde{H}_z = H_0\,C_X C_Y$$

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\,C_X S_Y$$

$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\,S_X C_Y$$

$$\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TE}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TE}}$$

TM mode *generator: $\tilde{E}_z$*

$$\tilde{E}_z = E_0\,S_X S_Y$$

$$\tilde{E}_x = -\frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right)E_0\,C_X S_Y$$

$$\tilde{E}_y = -\frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right)E_0\,S_X C_Y$$

$$\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\text{TM}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\text{TM}}$$

TEM (plane wave) for reference: $\tilde{E}_x = E_0 e^{-j\beta z}$, $\tilde{H}_y = \tilde{E}_x/\eta$, f_c N/A, $k = \omega\sqrt{\mu\varepsilon}$.

CUTOFF FREQUENCY

f_{mn} *m, n integers ≥ 0 (not both zero)*

$$f_{mn} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs ($a > b$)

Quantity	Formula
u_{p0}	$c/\sqrt{\varepsilon_r}$ (hollow: $u_{p0} = c$)
TE$_{10}$	$u_{p0}/(2a)$ (dominant)
TE$_{01}$	$u_{p0}/(2b)$
TE$_{20}$	$u_{p0}/a = 2f_{10}$
TE$_{11}$,TM$_{11}$	$\frac{u_{p0}}{2}\sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

ε_r FROM DISPERSION

Given ω, β, mode (m, n)

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]$$

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H_y

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}\ (\Omega) \qquad Z_{\text{TM}} = \eta\sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \qquad H_y = \frac{E_x}{Z_{\text{TE}}} \text{ (TE mode)}$$

$Z_{TE} > \eta$ always (denominator < 1); $Z_{TM} < \eta$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0}\sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$

$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \quad \Rightarrow \quad u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$

$$t = L/u_g \text{ (s)} \quad \text{travel time through guide of length } L$$

Higher modes have lower u_g $\Rightarrow$ they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

η_1, η_2 — imp. (Ω)
Γ — refl. coeff (unitless)
$\tau = 1 + \Gamma$ — trans. coeff
$k_i = \omega\sqrt{\varepsilon_{r,i}}/c$ (rad/m)
α_2, β_2 — if med 2 lossy
$z = 0$ — boundary plane
$\eta_0 = 120\pi \approx 377\,\Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
$n_i = \sqrt{\varepsilon_{r,i}}$ — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of $\hat{k}$+normal
$\perp$: $\mathbf{E} \perp$ this plane
$\parallel$: $\mathbf{E} \parallel$ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m); $a > b$
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
$Z_{\mathrm{TE/TM}}$ — wave imp. (Ω)
E_x, H_y — field comps (V/m, A/m)
$\eta = \eta_0/\sqrt{\varepsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\varepsilon_r}$ — bulk phase vel.
$u_p = u_{p0}/\sqrt{1-(f_c/f)^2}$
$u_g = u_{p0}\sqrt{1-(f_c/f)^2}$
$u_p u_g = u_{p0}^2$
$t = L/u_g$ — travel time
$c = 3 \times 10^8$ m/s

POWER & UNITS

$\%P_r = 100|\Gamma|^2$
$\%P_t = 100|\tau|^2\eta_1/\eta_2$
$\%P_r + \%P_t = 100$
σ (S/m), ε (F/m), μ (H/m)
$\varepsilon_0 = 8.85\times10^{-12}$ F/m
$\mu_0 = 4\pi\times10^{-7}$ H/m
$1/(36\pi)\times10^{-9} \approx \varepsilon_0$

DIPOLE TYPE BY LENGTH

Check first *compute $\lambda = c/f$, then l/λ*

l/λ	Type	Use
$< 1/50$	Hertzian (short)	Hertzian formula
$= 1/4$	Quarter-wave	Arbitrary formula
$= 1/2$	Half-wave	$D = 1.64$, simple
other	Arbitrary	Arbitrary formula

If $l/\lambda < 1/50$ stop and use Hertzian only — don't subst. small l into arbitrary formula.

HERTZIAN DIPOLE — $S(R, \theta)$

Power density *$l/\lambda < 1/50$; far field*

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2\theta \ (\mathrm{W/m^2})$$

$k = 2\pi/\lambda$ rad/m; I_0 peak current (A); l length (m); R obs. dist (m).

Max direction *$\theta = \pi/2$ (broadside)*

Total radiated power *integrate over 4π sr*

$$P_{\mathrm{rad}} = \frac{\eta_0 \pi}{3}\left(\frac{I_0 l}{\lambda}\right)^2 \ (\mathrm{W})$$

Directivity *Hertzian, exact*

$$D_{\mathrm{Hertz}} = 1.5 \quad (\text{i.e. } 1.76 \text{ dB})$$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at distance R *any l*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

Normalized pattern *$F(\theta) = S(\theta)/S_{\max}$, dimensionless*

Half-wave $l = \lambda/2$ *$\pi l/\lambda = \pi/2$*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}, \quad D = 1.64, \ R_{\mathrm{rad}} = 73\,\Omega$$

Quarter-wave $l = \lambda/4$ *$\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$*

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin\theta}\right]^2, \ S_0 = \frac{15 I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0\left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858\, S_0$$

BEAM PARAMETERS — HPBW, Ω_p, D

1. Normalize

$$F(\theta) = S(\theta)/S_{\max}$$

2. Half-power beamwidth (HPBW) *solve $F(\theta_{1/2}) = 0.5$*

$$\beta = 2\theta_{1/2} \quad (\text{symmetric beams})$$

example *$F(\theta) = e^{-20\theta^2} \Rightarrow \theta_{1/2} = \sqrt{-\ln 0.5/20} = 0.186$ rad $\Rightarrow \beta = 21.31°$*

3. Pattern solid angle *(sr)*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\, d\theta\, d\phi$$

If F has no ϕ dep.: $\Omega_p = 2\pi\int_0^{\pi} F(\theta)\sin\theta\, d\theta$.

4. Directivity

$$D = \frac{4\pi}{\Omega_p} \ (\text{unitless}) \qquad D_{\mathrm{dB}} = 10\log_{10} D$$

5. Gain *ξ = radiation efficiency, $0 < \xi \le 1$*

$$G = \xi D \qquad G_{\mathrm{dB}} = 10\log_{10} G$$

For lossless / ideal antennas: $G = D$.

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	R_{rad} (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

EFFECTIVE AREA A_e

Definition *m^2, antenna's receiving cross section*

$$A_e = \frac{\lambda^2 D}{4\pi}$$

Physical cross section *half-wave dipole, wire diameter d*

$$A_p = l\,d = \frac{\lambda}{2}\,d$$

Ratio *$A_e \gg A_p$ for thin wires (good!)*

Example: $f = 100$ MHz, $d = 2$ cm $\Rightarrow A_e/A_p \approx 39$.

POWER FLOW IDENTITIES

$$P_{\mathrm{rad}} = S_{\max}\,\Omega_p\,R^2 = \frac{4\pi R^2 S_{\max}}{D}$$

$$S_{\max} = \frac{P_{\mathrm{rad}}\,D}{4\pi R^2} \ (\mathrm{W/m^2})$$

For isotropic radiator: $S = P_{rad}/(4\pi R^2)$.

FRIIS TRANSMISSION FORMULA

Far-field link budget *P_t tx, P_r rx, dist R*

$$\boxed{P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t}$$

G_t, G_r linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.

Solve for R

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}}$$

Solve for G_r **given** A_e

$$G_r = \frac{4\pi A_{e,r}}{\lambda^2} \quad \Rightarrow \quad P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$$

Use linear G, not dB. Convert first: $G_{lin} = 10^{G_{dB}/10}$.

Free-space path loss $L_{fs} = (4\pi R/\lambda)^2$ so $P_r = G_t G_r P_t / L_{fs}$.

dB ↔ LINEAR

Power gain *factor → dB*

$$G_{\mathrm{dB}} = 10\log_{10} G_{\mathrm{lin}} \qquad G_{\mathrm{lin}} = 10^{G_{\mathrm{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

For E or amplitude use $20\log$ not $10\log$.

RADAR EQUATION (BONUS)

Monostatic radar *tx = rx, target cross section σ_t (m^2)*

$$P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}$$

Doppler shift *closing velocity v_r*

$$f_d = \frac{2 v_r}{\lambda} = \frac{2 v_r f}{c} \ (\mathrm{Hz})$$

Factor of 2 from round-trip path; v_r is the radial component toward radar.

LECTURE-11 BOUNDARY ↔ T-LINE

EM	T-line	Match
$\tilde{E}_i$	$\tilde{V}_1^+$	forward wave
$\tilde{E}_r$	$\tilde{V}_1^-$	reverse wave
η_i	Z_{0i}	line impedance
Γ	Γ	same formula
k_i	β_i	wavenumber

Bouncing-wave analogy: every t-line formula in Ch. 2 transfers verbatim with the substitutions above.

ANTENNA INTEGRATION TIPS

Many radiation integrals (e.g. Ω_p for $e^{-20\theta^2}\sin\theta$) have no closed form.

Use Wolfram / numerical integration. State this in the answer.

Common shortcut: if $F(\theta)$ is narrow ($\theta_{1/2} \ll 1$), approximate

$$\Omega_p \approx \theta_{1/2}^2\,\pi/(4\ln 2) \quad (\text{Gaussian beam})$$

DIPOLE

- l — antenna length (m)
- $\lambda = c/f$ — wavelength (m)
- I_0 — peak current (A)
- $\eta_0 = 120\pi \approx 377\,\Omega$
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- R_{rad} — radiation res. (Ω)
- R — obs. dist (m)

RADIATION

- $S(R,\theta)$ — power density ($\mathrm{W/m^2}$)
- $S_{\max}$ — peak power density
- $F(\theta) = S/S_{\max}$ — normalized
- Ω_p — pattern solid angle (sr)
- β — HPBW (rad or deg)
- D — directivity (unitless)
- P_{rad} — total rad. power (W)

ANTENNA

- $A_e = \lambda^2 D/(4\pi)$ — effective area
- A_p — physical cross sec. ($\mathrm{m^2}$)
- ξ — rad. efficiency ($0 < \xi \le 1$)
- $G = \xi D$ — gain (unitless)
- $G_{\mathrm{dB}} = 10\log G$
- Half-wave: $D = 1.64$, $R_{\mathrm{rad}} = 73\,\Omega$
- Hertzian: $D = 1.5$

FRIIS / LINK

- P_t — tx power (W)
- P_r — rx power (W)
- G_t, G_r — linear gains
- R — range (m)
- $L_{\mathrm{fs}} = (4\pi R/\lambda)^2$ — path loss
- 3 dB→2; 10 dB→10
- $G_{\mathrm{lin}} = 10^{G_{\mathrm{dB}}/10}$

RADAR

- σ_t — RCS ($\mathrm{m^2}$)
- f_d — Doppler shift (Hz)
- v_r — radial vel. (m/s)
- $P_r \propto 1/R^4$ (monostatic)
- $f_d = 2v_r/\lambda$ (round trip)
- $c = 3\times10^8$ m/s
- $f_d > 0$: target approaching

TEXTBOOK REFERENCE TABLES

Table 7-1 — Lossy media: α, β, η_c, u_p, λ by regime

Table 7-1 Expressions for α, β, η_c, u_p, and λ for various types of media.

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\epsilon''/\epsilon' \ll 1$)	Good Conductor ($\epsilon''/\epsilon' \gg 1$)	Units
$\alpha =$	$\omega \left[\frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} - 1 \right] \right]^{1/2}$	0	$\frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega \left[\frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} + 1 \right] \right]^{1/2}$	$\omega\sqrt{\mu\epsilon}$	$\omega\sqrt{\mu\epsilon}$	$\sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\epsilon'}} \left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{-1/2}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\frac{\mu}{\epsilon}}$	$(1 + j)\frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω/β	$1/\sqrt{\mu\epsilon}$	$1/\sqrt{\mu\epsilon}$	$\sqrt{4\pi f/\mu\sigma}$	(m/s)
$\lambda =$	$2\pi/\beta = u_p/f$	u_p/f	u_p/f	u_p/f	(m)

Notes: $\epsilon' = \epsilon$; $\epsilon'' = \sigma/\omega$; in free space, $\epsilon = \epsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\epsilon''/\epsilon' = \sigma/\omega\epsilon < 0.01$ and a good conducting medium if $\epsilon''/\epsilon' > 100$.

Table 7-2 — dB $\leftrightarrow$ linear power ratios

Table 7-2 Power ratios in natural numbers and in decibels.

G	G [dB]
10^x	$10x$ dB
4	6 dB
2	3 dB
1	0 dB
0.5	-3 dB
0.25	-6 dB
0.1	-10 dB
10^{-3}	-30 dB

Table 8-1 — Normal incidence plane wave $\leftrightarrow$ transmission line analogy

Table 8-1 Analogy between plane-wave equations for normal incidence and transmission-line equations, both under lossless conditions.

Plane Wave [Fig. 8-4(a)]		Transmission Line [Fig. 8-4(b)]	
$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^{\mathrm{i}} (e^{-jk_1 z} + \Gamma e^{jk_1 z})$	(8.11a)	$\widetilde{V}_1(z) = V_0^+ (e^{-j\beta_1 z} + \Gamma e^{j\beta_1 z})$	(8.11b)
$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}}\, \frac{E_0^{\mathrm{i}}}{\eta_1} (e^{-jk_1 z} - \Gamma e^{jk_1 z})$	(8.12a)	$\tilde{I}_1(z) = \frac{V_0^+}{Z_{01}} (e^{-j\beta_1 z} - \Gamma e^{j\beta_1 z})$	(8.12b)
$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^{\mathrm{i}} e^{-jk_2 z}$	(8.13a)	$\widetilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2 z}$	(8.13b)
$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau\, \frac{E_0^{\mathrm{i}}}{\eta_2}\, e^{-jk_2 z}$	(8.14a)	$\tilde{I}_2(z) = \tau\, \frac{V_0^+}{Z_{02}}\, e^{-j\beta_2 z}$	(8.14b)
$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$		$\Gamma = (Z_{02} - Z_{01})/(Z_{02} + Z_{01})$	
$\tau = 1 + \Gamma$		$\tau = 1 + \Gamma$	
$k_1 = \omega\sqrt{\mu_1\epsilon_1}\,, \quad k_2 = \omega\sqrt{\mu_2\epsilon_2}$		$\beta_1 = \omega\sqrt{\mu_1\epsilon_1}\,, \quad \beta_2 = \omega\sqrt{\mu_2\epsilon_2}$	
$\eta_1 = \sqrt{\mu_1/\epsilon_1}\,, \quad \eta_2 = \sqrt{\mu_2/\epsilon_2}$		Z_{01} and Z_{02} depend on transmission-line parameters	

Table 8-2 — Γ, τ, R, T for normal / $\perp$ / $\parallel$ polarization

Table 8-2 **Expressions for Γ, τ, R, and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2. Angles θ_i and θ_t are the angles of incidence and transmission, respectively.**

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \dfrac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_\perp = \dfrac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\Gamma_\parallel = \dfrac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Transmission coefficient	$\tau = \dfrac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_\perp = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\tau_\parallel = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel) \dfrac{\cos\theta_i}{\cos\theta_t}$
Reflectivity	$R = \lvert\Gamma\rvert^2$	$R_\perp = \lvert\Gamma_\perp\rvert^2$	$R_\parallel = \lvert\Gamma_\parallel\rvert^2$
Transmissivity	$T = \lvert\tau\rvert^2 \left(\dfrac{\eta_1}{\eta_2}\right)$	$T_\perp = \lvert\tau_\perp\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$	$T_\parallel = \lvert\tau_\parallel\rvert^2 \dfrac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$
Relation of R to T	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: (1) $\sin\theta_t = \sqrt{\mu_1\epsilon_1/\mu_2\epsilon_2}\,\sin\theta_i$; (2) $\eta_1 = \sqrt{\mu_1/\epsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2/\epsilon_2}$; (4) for nonmagnetic media, $\eta_2/\eta_1 = n_1/n_2$.

Table 8-3 — TE / TM / TEM full field components in rectangular waveguide

Table 8-3 **Wave properties for TE and TM modes in a rectangular waveguide with dimensions $a \times b$, filled with a dielectric material with constitutive parameters ϵ and μ. The TEM case, shown for reference, pertains to plane-wave propagation in an unbounded medium.**

Rectangular Waveguides		Plane Wave
TE Modes	TM Modes	TEM Mode
$\widetilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\, e^{-j\beta z}$	$\widetilde{E}_x = \frac{-j\beta}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\, e^{-j\beta z}$	$\widetilde{E}_x = E_{x0} e^{-j\beta z}$
$\widetilde{E}_y = \frac{-j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\, e^{-j\beta z}$	$\widetilde{E}_y = \frac{-j\beta}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\, e^{-j\beta z}$	$\widetilde{E}_y = E_{y0} e^{-j\beta z}$
$\widetilde{E}_z = 0$	$\widetilde{E}_z = E_0 \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)\, e^{-j\beta z}$	$\widetilde{E}_z = 0$
$\widetilde{H}_x = -\widetilde{E}_y / Z_{TE}$	$\widetilde{H}_x = -\widetilde{E}_y / Z_{TM}$	$\widetilde{H}_x = -\widetilde{E}_y / \eta$
$\widetilde{H}_y = \widetilde{E}_x / Z_{TE}$	$\widetilde{H}_y = \widetilde{E}_x / Z_{TM}$	$\widetilde{H}_y = \widetilde{E}_x / \eta$
$\widetilde{H}_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)\, e^{-j\beta z}$	$\widetilde{H}_z = 0$	$\widetilde{H}_z = 0$
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	$Z_{TM} = \eta\sqrt{1 - (f_c/f)^2}$	$\eta = \sqrt{\mu/\epsilon}$
Properties Common to TE and TM Modes		
$f_c = \dfrac{u_{p_0}}{2}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2}$		$f_c =$ not applicable
$\beta = k\sqrt{1 - (f_c/f)^2}$		$k = \omega\sqrt{\mu\epsilon}$
$u_p = \dfrac{\omega}{\beta} = u_{p_0} / \sqrt{1 - (f_c/f)^2}$		$u_{p_0} = 1/\sqrt{\mu\epsilon}$